

Part I: Objective questions [60 min; $10 \times 6 = 60$ points]

For Multiple Choice Questions (MCQs), one or more options may be correct (tick/select the correct option(s)). For True/False (T/F), write T or F. For Fill In The Blanks (FITBs), write the most precise answer. Steps/details not required.

1. **IPL Wicket-Prediction Classifier.** A franchise builds a binary classifier that, ball by ball, predicts whether the next delivery will result in a wicket. On a test set of 1,000 balls (of which 15 actually produce wickets), the model flags 20 balls as "wicket-likely" and gets 12 of the actual wickets correct. [2+2+2]

1.5 a) The confusion matrix entries are: $TP = 12$, $FP = 8$, $FN = 3$,
 $TN = 1000 - (TP + FP + FN) = 1000 - 26 = 974$ → formula correct

2 b) The precision of the wicket class equals $\frac{12}{20}$ and the recall equals $\frac{12}{15}$. correct formula

c) (MCQ) The franchise's analyst declares the model "96% accurate". Which are valid criticisms?

(i) Accuracy is misleading because of severe class imbalance.

2 (ii) A trivial classifier predicting "no-wicket" always would already exceed 95% accuracy.

(iii) F1-score / AP would be a more honest summary for this skewed problem.

(iv) Confusion matrices can never be used when classes are imbalanced.

2. **Linear Regression: Predicting Strike Rate.** A linear regression prediction $\hat{y}_i = \mathbf{w}^T \mathbf{x}_i$ is fit to predict batter strike-rate from features such as balls faced and boundary count, by minimising MSE over N samples, data matrix X . [2+2+2]

2 a) (T/F) The closed-form solution $\mathbf{w}^* = (X^T X)^{-1} X^T \mathbf{y}$ exists uniquely even when $X^T X$ is singular. No, Inverse doesn't exist.

2 b) The gradient of the MSE objective (∇J) with respect to $\mathbf{w}$ equals, $\nabla J = -\frac{2}{N} \sum (y_i - \mathbf{w}^T \mathbf{x}_i) \mathbf{x}_i$.

c) (MCQ) If the team adds an ℓ_2 regulariser $\lambda \|\mathbf{w}\|_2^2$ (Ridge), then:

2 (i) The closed-form becomes $(X^T X + \lambda I)^{-1} X^T \mathbf{y}$.

(ii) Some weights are forced to be exactly zero.

(iii) Numerical stability of the inverse improves.

(iv) Ridge generally yields sparser solutions than Lasso.

$$X = \left[\begin{matrix} & 10^4 \\ & \\ & \end{matrix} \right] \} 400 = 400 \times 10^4$$

3. PCA / Eigen-Faces. A research group has $N = 400$ frontal face images of size 100×100 pixels (mean-subtracted) stacked as rows of X . [2+2+2]

2 a) The covariance matrix $\Sigma = \frac{1}{N} X^T X$ has size $\frac{10^4}{400} \times \frac{10^4}{400}$, and its rank is at most 400. (rank = min (concepts, features, samples))

2 b) (T/F) The non-zero eigenvalues of $X^T X$ and $X X^T$ are identical, so it is computationally cheaper here to first eigen-decompose $X X^T$ and then map eigenvectors back. True

c) (MCQ) About PCA:

✓ (i) It is an *unsupervised* linear method.

✓ (ii) Its principal directions maximise variance and, equivalently, minimise reconstruction error.

2 (iii) It is identical to LDA/Fisher when the class labels happen to be balanced.

✓ (iv) The covariance matrix is symmetric and positive semi-definite.

4. Bayesian Decision in a Classroom. A professor classifies each enrolled student as "will attend tomorrow" (ω_1) or "will skip" (ω_2). Past records suggest $P(\omega_1) = 0.7$, $P(\omega_2) = 0.3$, with class-conditional likelihoods $p(x | \omega_1) = 0.4$ and $p(x | \omega_2) = 0.6$ for the observed feature x . [4 + 2]

4 a) The posterior using Bayes rule $P(\omega_1 | x)$ is ~~0.5~~ 14/23.

2 b) (T/F) The Bayes classifier achieves the *minimum possible* probability of error among all classifiers. True

$$P(\omega_1) = 0.7$$

$$P(y=1|x) = \frac{P(x|y=1) \cdot P(y=1)}{P(x)} = \frac{1}{1+e^{-x}}$$

5. Naive Bayes & Logistic Regression. A spam filter for college email uses Naive Bayes over d binary word-presence features. [2 + 2 + 2]

2 a) (T/F) Naive Bayes assumes the features are independent. **True**

b) (FITB) For logistic regression with the convention $y \in \{0, 1\}$, the model probability is $p(y=1|x)$, and a frequently-used loss is the negative log-likelihood loss. **MLE** (a.k.a. **MAP** (maximum a posteriori))

Given x , p can be either 0 or 1. Hence $P(y=1|x) = P(x|y=1) \cdot P(y=1) = \frac{1}{1+e^{-x}}$

c) (MCQ) Picking the squared-error loss $\frac{1}{2N} \sum_i (\sigma(w^T x_i) - y_i)^2$ for logistic regression is a poor choice because:

(i) The objective is non-convex in w .

(ii) Gradients vanish in the saturated regions of the sigmoid, slowing learning.

(iii) The cross-entropy / log-loss is the natural MLE for a binary classification model.

Check (iv) Squared error always violates Bayes optimality.

$$\frac{\partial J}{\partial w} = \frac{1}{2N} \sum_i [(\sigma(w^T x_i) - y_i)] \cdot \frac{\partial (\sigma(w^T x_i) - y_i)}{\partial w} \rightarrow \sigma(x) = \frac{1}{1+e^{-x}} \rightarrow \text{non convex}$$

6. SVMs and Kernels. [3 + 3]

a) (MCQ) For the standard hard-margin SVM $\min \frac{1}{2} \|w\|^2$ subject to $y_i(w^T x_i + b) \geq 1$:

(i) Only the support vectors have non-zero dual variables α_i .

(ii) The geometric margin equals $2/\|w\|$.

(iii) The dual objective depends on data only through inner products $x_i^T x_j$, enabling the kernel trick.

(iv) Replacing the constraint by the hinge loss $\max(0, 1 - y_i f(x_i))$ recovers exactly the same primal problem in the soft-margin case.

3 b) (T/F) For a soft-margin SVM with parameter C , increasing C reduces the tolerance for margin violations and tends to produce a classifier with a narrower margin. **True**

7. From SLP to MLP. [3 + 3]

a) (MCQ) A Single-Layer Perceptron:

- ☒ (i) Is guaranteed to converge in a finite number of steps if the data are linearly separable.
- (ii) Can solve the 2-input XOR problem with no kernel expansion. ~~X~~
- ☐ (iii) Always converges (with appropriate η) when used as a linear regressor under the delta rule, even with noisy / non-separable data.
- ☒ (iv) Has expressive power equivalent to a sigmoid MLP with one hidden layer.

b) (FITB) Consider an MLP with 3 inputs, one hidden layer of 2 sigmoid units, and 1 linear output unit. Every hidden and output unit has a bias. The total number of learnable scalar parameters is _____.

3
$$z_1 = w_{11}x_1 + w_{12}x_2 + w_{13}x_3 + b_1 \rightarrow h_1 = \sigma(z_1) \Rightarrow 4$$

$$z_2 = w_{21}x_1 + w_{22}x_2 + w_{23}x_3 + b_2 \rightarrow h_2 = \sigma(z_2) \Rightarrow 4$$

$$v = v_1h_1 + v_2h_2 + b_3 \Rightarrow 3$$

(11) ans.

8. Optimisers, BatchNorm and Dropout. [2 + 2 + 2]

a) (T/F) Adam combines momentum (a running first moment of the gradient) with an adaptive per-parameter learning rate (a running second moment), and applies bias-correction to both estimates.

b) (MCQ) The vanishing-gradient problem in deep networks can be mitigated by:

- ☒ (i) Using ReLU-family activations instead of saturating sigmoids.
- ☒ (ii) Using residual / skip connections.
- ☒ (iii) Batch normalisation between linear and non-linear layers.
- (iv) Initialising every weight to zero.

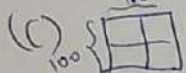
c) (MCQ) Which of the following describes the primary original motivation proposed for incorporating Batch Normalization into deep network training?

- (i) To stabilize training distributions by reducing Internal Covariate Shift.
- ☒ (ii) To drastically decrease the computational floating-point operations (FLOPs) of the network.
- (iii) To convert arbitrary non-linear feature maps into strictly linear mappings.
- (iv) To compress the final serialized model size for mobile-edge deployment.

9. 2D Convolution Arithmetic. A document-structure recogniser takes a binarised page of size $224 \times 224 \times 1$ as input. $[2 + 2 + 2]$

- a) (FITB) The first conv layer applies 32 filters of size 7×7 , with stride 2 and padding 3.
 Spatial output size = $\frac{100}{2} \times \frac{100}{2}$, number of output channels = ~~no. of input channels~~ filters = 32
- b) (FITB) The number of learnable parameters in this conv layer (including biases) is 72.
 learnable parameters = input channel + output channel + filter size = $1 + 32 + 7 = 32 + 72 = 72$
- c) (FITB) Its output is followed by a 2×2 max-pool with stride 2. The resulting feature-map shape is $\frac{25}{2} \times \frac{25}{2} \times 32$.
 bias in output state

(a)
$$W_{out} = \frac{W_{in} - K + 2P}{S} + 1 = \frac{224 - 7 + 2(3)}{2} + 1 = \frac{224 - 7 + 6}{2} + 1 = \frac{223}{2} + 1 = 111.5 + 1 = 112.5$$



10. Consider a multihead self attention layer with 8 heads. Input as well as output are 1000 vectors (N) of 100 dimension (d). Query, Key and Values are 50 dimension. $[2 + 2 + 2]$ (1000×100)

We know that there are multiple matrices W_Q , W_K , W_V and W_O matrices.

- How many total matrices are there?
- Write the dimensionality of the matrices.
- total number of learnable parameters.

exist. so $8 \times 3 = 24$
 heads = $24 + 1 = 25$

Part II: Answer three out of four of the following questions [60 min;
 $3 \times 20 = 60$ points]

1. Backpropagation / Gradient Descent on a Single Neuron. Consider one sigmoid neuron $o = \phi(w_1x_1 + w_2x_2 + b)$ with $\phi(z) = 1/(1 + e^{-z})$ and squared loss $L = \frac{1}{2}(o - y)^2$. Current parameters: $w_1 = 1, w_2 = -1, b = 0$. Training sample: $\mathbf{x} = [1, 1]^T, y = 1$. Learning rate $\eta = 1$. [3 + 10 + 8]

- Derive $\phi'(z)$ and express in terms of $\phi(z)$.
- Derive expressions using chain rule for $\partial L / \partial w_1, \partial L / \partial w_2$, and $\partial L / \partial b$, and evaluate them numerically.
- Perform one Gradient Descent (GD) update for the training sample provided starting from the given values and report the new (w_1, w_2, b) .

2. **Bayesian Classification in 1D** We have two classes A and B each with $p(x | x \in A) = p(x | x \in B) = \mathcal{N}(\mu, \sigma^2)$. We also know $\mu_A = 100$ and $\mu_B = 200$. $\sigma_A^2 = \sigma_B^2 = 100$. We know prior probabilities of class A and B as 80% and 20%. [5 + 10 + 5]

- a) Write expression for $p(x|x \in A)$ and $p(x|x \in B)$. Evaluate these two numerically for $x = 150$. Should they sum upto 1.0?
- b) Write Bayes theorem. Classify $x = 149$; $x = 150$ and $x = 151$ into A or B using Bayes classification rule. If you have computed pairs of probabilities to compare, should they sum up to 1.0?
- c) Numerically compute the posterior probability $P(x \in A|x)$ and $P(x \in B|x)$. Should they sum upto 1.0?

3. Eigen decomposition and SVD. Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

[6 + 8 + 6]

- a) Find both eigenvalues of A and the corresponding unit-norm eigenvectors.
- b) Given that you now know eigen vectors and eigen values of A , Write the SVD $A = UDV^T$ explicitly as 3 matrices U , D and V^T .
- c) Compute the best rank-1 approximation A_1 (as discussed in the class) $\min_{A_1} \|A - A_1\|^2$.

[4 + 6 + 10]

4. 1D K-Means.

- a) Write the K-Means objective $L = J(\{C_k\}, \{\mu_k\})$ being minimised.
- b) Let L^* be the optimal loss when the algorithm converges. How will L^* change if we change K ? Plot a potential graph with K on X-axis and L^* on Y-axis. Can/how we use this graph to find the 'right' K for a dataset?
- c) Consider K Means with a simple one dimensional data ($d = 1$, $N=6$ samples). Data: $\{1, 2, 3, 8, 9, 10\}$. Assume $K = 2$ with initial means $\mu_1^{(0)} = 1, \mu_2^{(0)} = 2$. Find the subsets/clusters C_k and their means μ_k . Run iterations until convergence. For each iteration, list (i) cluster assignments, (ii) updated means, and (iii) the value of J . Stop when assignments do not change.

$\sum_{i \in C_k} \|x_i - \mu_k\|^2$

Part III: Answer three out of four of the following questions [60 min; $3 \times 20 = 60$ points]

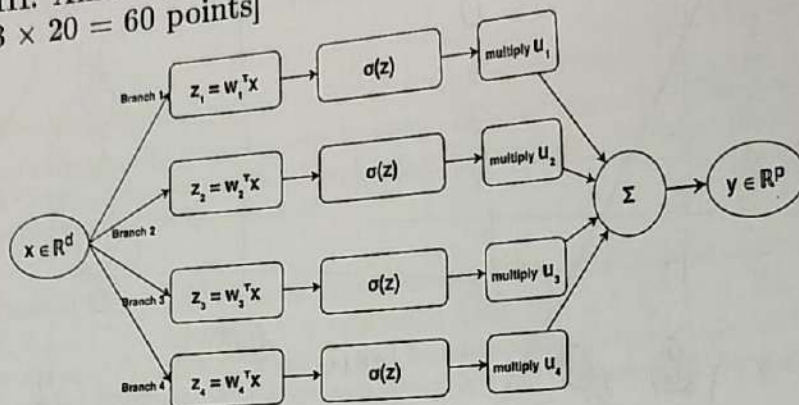


Figure 1: Parallel / Wider Neural Network with 4 Branches

1. **Parallel / Wider Neural Network with 4 Branches.** The input $x \in \mathbb{R}^d$ is fed in parallel to four branches (see Figure above). Branch $k \in \{1, 2, 3, 4\}$ has a weight vector $w_k \in \mathbb{R}^d$ and an output projection (learnable parameter) vector $u_k \in \mathbb{R}^p$. It first computes the scalar $z_k = w_k^T x$ and $\alpha_k = \sigma(z_k) = \sigma(w_k^T x)$ and then contributes $\alpha_k u_k$ to the final output.

[6 + 7 + 7]

- Write an expression or python code to compute y given x .
- If $d = 1$, compute $\partial y / \partial x$
- For general p (and d), compute $\partial y / \partial z_2$

2. **Kernel Matrix.** Let $\kappa(\mathbf{p}, \mathbf{q}) = \phi(\mathbf{p})^T \phi(\mathbf{q})$. For samples $\{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ define $K_{ij} = \kappa(\mathbf{x}_i, \mathbf{x}_j)$.
[2 + 8 + 10]
- a) Show that K is symmetric.
 - b) Show that K is positive semi-definite by proving $\mathbf{v}^T K \mathbf{v} \geq 0$ for all $\mathbf{v} \in \mathbb{R}^N$.
 - c) For $\mathbf{p}, \mathbf{q} \in \mathbb{R}^2$ and $\kappa(\mathbf{p}, \mathbf{q}) = (\mathbf{p}^T \mathbf{q})^3$, find an explicit feature map $\phi(\cdot)$ such that $\kappa(\mathbf{p}, \mathbf{q}) = \phi(\mathbf{p})^T \phi(\mathbf{q})$.

(a) for a matrix to be symmetric $A = A^T$

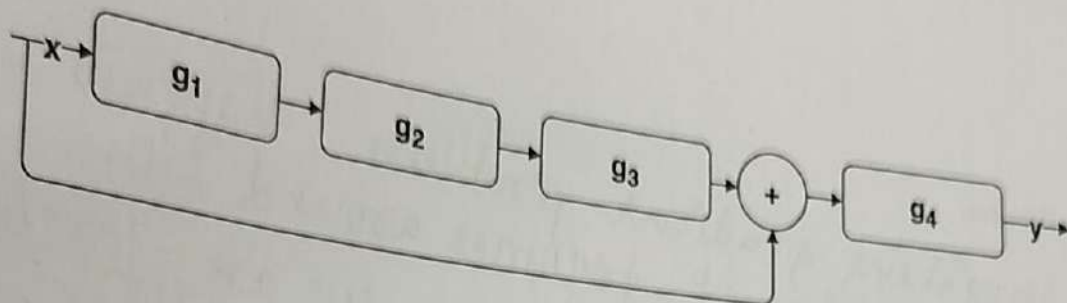


Figure 2: Residual Block

3. **Backprop Through a Residual Block.** Consider a residual block (see Figure above). Input is x , each g_i is a function whose gradient is g'_i . [12 + 4 + 4]

- a) Write $\partial y / \partial x$ using chain rule.
- b) How do residual connections help with vanishing gradient problem? Give a single line answer.
- c) How does ReLU help with vanishing gradient problem?

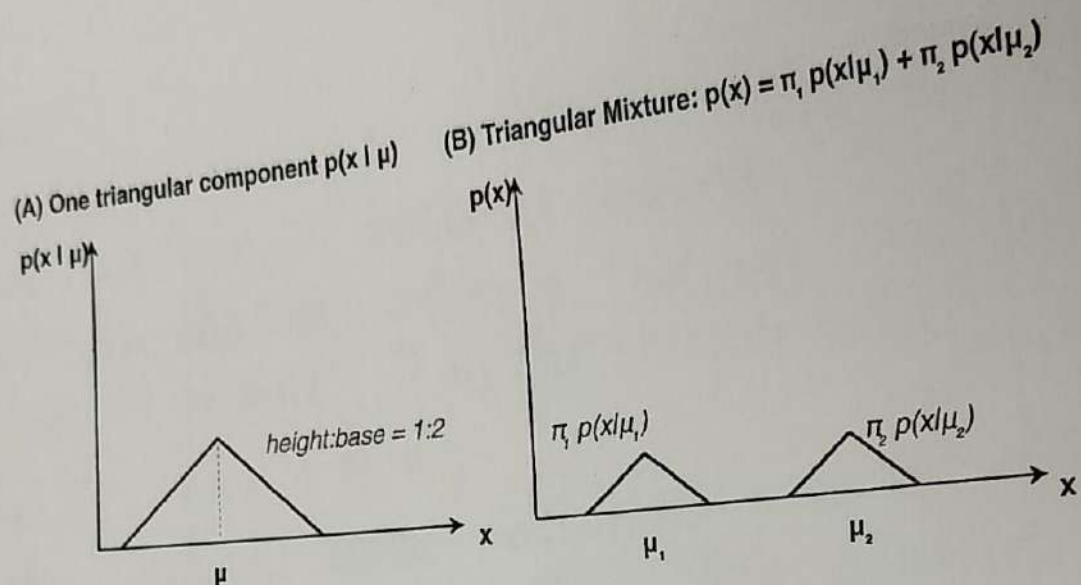


Figure 3: Triangular Mixture Model (TMM)

4. Triangular Mixture Model (TMM). You know the Gaussian Mixture Model (GMM) Algorithm. Gaussians have a long tail and also a complex expression (with exponent etc.) Let us assume data is univariate. Assume we simplify the gaussian as a simple triangle symmetric around mean, with base:height as 1:2. (see Figure above). The probability is zero outside its base.

- Write an expression (or python code) for computing $p(x | \mu)$.
(Hint: (i) area under the curve = 1.0 (ii) since the function is symmetric, use of $|x - \mu|$ could simplify.)
write $p(x | \mu)$ as a piecewise-linear function of $|x - \mu|$, and state the support.
- Recall GMM algorithm and write steps with appropriate equations. convergence criterion.
- In a similar way, write your own TMM algorithm.
- Why TMM is better/worse compared to GMM? (write 2-3 advantages and disadvantages for both)